

Understanding Partisan Bias in Judgments of Misinformation: Identity Protection versus Differential Knowledge

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People over-accept information that supports their identity and under-accept information that opposes their identity—a phenomenon known as partisan bias. While partisan-bias effects in judgments of misinformation are robust and pervasive, there is ongoing debate about whether partisan-bias effects arise from identity-protective motivated reasoning or differential knowledge of identity-congenial versus identity-uncongenial information. Prior empirical work has been unable to differentiate the two accounts due to a reliance on groups with pre-existing differences in knowledge (e.g., Democrats and Republicans). The current research addresses this issue by using randomly assigned rather than pre-existing identities. Across two experiments ($N_{total} = 1,411$), participants showed lower thresholds for accepting information that is congenial versus uncongenial to a randomly assigned identity, despite no differences in prior knowledge. These results support theories that emphasize identity protection as a factor underlying partisan bias in the acceptance of misinformation, with important practical implications for misinformation interventions.

Keywords: Identity Protection; Misinformation; Motivated Reasoning; Partisan Bias; Prior Knowledge; Signal Detection; Truth Judgment

Given the vast amount of misinformation on the internet and other media, people are commonly tasked with deciding what is true and what is false. Although people are quite good at discerning true from false information (Gawronski et al., 2025; Pennycook & Rand, 2021a; Pfänder & Altay, 2025), they do make systematic errors. In particular, people tend to accept information that is congenial to their personal beliefs and to reject information that is uncongenial to their personal beliefs, which can lead them to mistakenly judge false information as true if it supports their views and identities, and to mistakenly judge true information as false if it opposes their views and identities (Gawronski et al., 2025). This tendency—which we refer to as partisan bias—is one of the most robust and pervasive factors in judgments of misinformation (Batailler et al., 2022; Gawronski et al., 2023; Nahon et al., 2024).

While the existence of partisan bias in judgments and information processing is supported by a large body of evidence, the mechanisms underlying partisan bias are subject of ongoing debate (Ditto et al., 2025). Some have argued that partisan bias is driven in part by motivated reasoning (Celniker & Ditto, 2024; Ditto et al., 2025; Kahan, 2016; Peterson & Iyengar, 2020; Van Bavel et al., 2024). Yet, others have pointed out that partisan bias can be explained in purely cognitive terms as the product of differential knowledge (Baron & Jost, 2019; Druckman & McGrath, 2019; Pennycook & Rand, 2021b; Tappin et al., 2020). The current research aims to advance this debate by testing whether people show partisan bias in judgments of true and false information even without differences in prior knowledge.

Identity Protection versus Differential Knowledge

At the heart of the debate about the underpinnings of partisan bias is the concept of motivated reasoning, or the tendency to arrive at conclusions one wants to arrive at (Kunda, 1990). While there are many motivations that might influence judgments (Briñol & Petty, 2005), one

particularly relevant motivation is the need for identity protection. According to theories emphasizing identity protection, partisans over-accept information that supports their side and under-accept information that opposes their side because they are motivated to support and protect their partisan identity (Van Bavel et al., 2024).

At first glance, partisan bias in truth judgments may appear to be a clear case of identity-protective motivated reasoning. However, some researchers have pointed out that partisan bias can also be explained in purely cognitive terms (Baron & Jost, 2019; Druckman & McGrath, 2019; Pennycook & Rand, 2021b; Tappin et al., 2020). According to this purely cognitive account, partisan bias occurs because people typically have more knowledge of information that supports their partisan identity compared to information that opposes their partisan identity (Pennycook & Rand, 2021b). Such knowledge differences are often rooted in selective exposure, in that people tend to seek out information that is congenial to their views and to avoid information that is uncongenial to their views (Hart et al., 2009; Stroud, 2010). Because information is more likely judged as true if it is consistent rather than inconsistent with one's prior knowledge (Brashier & Marsh, 2020; Gawronski, 2012), greater knowledge of identity-congenial than identity-uncongenial information should increase the likelihood that new information is judged as true when it supports rather than when it opposes one's partisan identity.

Given the strong empirical support for differential knowledge across partisan groups, it is worth noting that proponents of the identity-protection account do not deny the role of knowledge-driven effects. Rather, the critical question under debate concerns the presumed role of identity-protective motivated reasoning as a factor underlying partisan bias (Ditto et al., 2025).

Empirically Testing Identity Protection

Providing evidence for identity-protective motivated reasoning that cannot be explained in terms of differential knowledge has proven notoriously challenging (Ditto, 2010; Druckman & McGrath, 2019; Tetlock & Levi, 1982). One line of research that has attempted to provide evidence for identity-protective motivated reasoning has used matched-information paradigms (see Ditto et al., 2025). In matched-information paradigms, partisans are asked to judge information that differs only in terms of its slant. Despite the information being otherwise identical, participants tend to judge identity-congenial information more favorably than identity-uncongenial information (e.g., Cohen, 2003; Lord et al., 1979).

Although results of matched-information studies are often interpreted as evidence for motivated reasoning, closer causal analysis demonstrates that these studies are incapable of differentiating between motivated reasoning and differential knowledge (Ditto et al., 2025; Tappin et al., 2020). This ambiguity is rooted in the fact that any findings involving pre-existing groups (e.g., Democrats and Republicans) can be explained by systematic differences in prior knowledge between these groups. Thus, addressing the debate requires alternative methods that convincingly rule out knowledge differences as a driving force. While some studies have attempted to address the problem by statistically controlling for prior knowledge (Celniker & Ditto, 2024), no prior research has ruled out differential knowledge by experimentally manipulating group membership.

The Current Research

The current research aims to provide a more compelling test of identity protection as a factor underlying partisan bias in judgments of true and false information by using randomly assigned instead of pre-existing identities. In two experiments, we randomly assigned participants to one of two teams (experimental conditions) or to no team (control condition) and asked them to judge the veracity of true and false information that favored one team over the other. By using random assignment to groups, we eliminate the possibility of pre-existing knowledge differences, thereby isolating identity protection as a potential driver of partisan bias. While the identity-protection account predicts that participants should be more likely to accept information that favors their randomly assigned ingroup compared to information that favors their randomly assigned outgroup, the differential-knowledge account does not predict partisan bias for randomly assigned groups.

Experiment 1

Method

Participants and Design

Experiment 1 utilized a 3 (Group Assignment: Team UK vs. Team France vs. No Team) $\times$ 2 (Statement

Accuracy: true vs. false) $\times$ 2 (Statement Slant: pro-UK vs. pro-France) mixed design with the first factor varying between-subjects and the other two factors varying within-subjects. We aimed to recruit a total of 600 participants. Sensitivity analyses using GPower 3.1.9.6 (Faul et al., 2007) demonstrated that a sample of 600 provides 80% power to detect a small effect of $f = 0.075$ for the predicted interaction between Group Assignment and Statement Slant, assuming a correlation between measures of $r = .30$ and using a nonsphericity correction of $\epsilon = 1$.

Participants were recruited on Prolific Academic. Participation was restricted to Prolific workers who (1) are U.S. citizens currently residing in the U.S., (2) are fluent in English, (3) have completed at least 100 previous assignments, and (4) had an approval rating of 95% or higher on previous submissions. Completion of the study took approximately 15 minutes and participants were compensated with US-\$3.

A total of 601 participants completed the study (one participant completed the study but did not request compensation). Following our preregistered exclusion criteria, participants were excluded if they failed an instructional attention check ($n_{\text{Team UK}} = 9$, $n_{\text{Team France}} = 14$, $n_{\text{No Team}} = 15$; Oppenheimer et al., 2009). Exclusions did not significantly differ across conditions, $\chi^2(2) = 1.65$, $p = .438$. After applying this exclusion criterion, our final sample consisted of 563 participants (190 Team UK; 185 Team France; 188 No Team). Demographic information is provided in Table 1.

Procedure

Participants from the U.S. were randomly assigned to either a *Team UK* condition or a *Team France* condition (experimental conditions) or a *No Team* condition (control condition). The UK and France were used for the two teams given that the two countries are relatively similar, have a history of conflict, and are well-known to U.S. participants. To bolster the effectiveness of the group manipulation, all participants were informed that the study concerned “differences between people from countries that have experienced conflict” and were provided a brief description of the UK and France as being in conflict with one another, along with images of the British flag and the French flag. The position of the two flags (left vs. right) was counterbalanced across participants. For the assignment to the two teams, participants completed a bogus personality assessment (i.e., a short version of the Big-Five; Soto & John, 2017), ostensibly aimed at determining with which country their personality was most closely aligned. Previous work within the minimal-groups literature has found this procedure to be an effective method for inducing identification with a novel group (Pinter & Greenwald, 2011). After completing the bogus personality assessment, participants in the experimental conditions were shown a loading screen, followed by their team

assignment which stated: “Based on the results of your personality test, you are on... [Team UK/Team France]”. For participants in the control condition, the study proceeded to the next part after the personality test without bogus feedback linking them to one of the two countries.

Following the bogus personality test, all participants completed a series of demographic questions asking them about their gender, age, ethnicity, race, and education. Next, all participants completed a series of questions assessing their identification with the UK and France (Pinter & Greenwald, 2011). These items included three questions per country (six total) asking about their identification with (“I identify with [the UK/France]”), attachment to (“I feel attached to the UK/France”), and attitude toward (“I like [the UK/France]”) each country. Responses were measured with 7-point rating scales (1 = *Strongly disagree*; 7 = *Strongly agree*). The three questions were blocked by country and presented in the same fixed order for each country. The order of the two countries (UK first vs. France first) was counterbalanced across participants.

For the truth-judgment task, participants read a total of 60 statements and responded either *True* or *False* to the question: *To the best of your knowledge, is the claim in this statement true or false?* The statements varied in terms of their veracity (true vs. false) and slant (pro-UK vs. pro-France). Examples of the statements used include: *In terms of GDP, the UK has a stronger economy than France* (pro-UK true) and *France ranked higher than the UK on a measure of political stability in 2022* (pro-France false). Statements were pilot tested to ensure that the slant of each statement matched the respective condition and that the average slant was of equal strength across conditions.¹ Statements were also balanced for valence such that all statement conditions contained an equal number of positive (e.g., pro-France in the pro-France statement condition), negative (e.g., anti-UK in the pro-France statement condition), and comparative (e.g., France better than the UK in the pro-France statement condition) statements. For further details on the selection and pilot testing of statements, see Supplemental Materials.

Data Analytic Approach

Following prior research on partisan bias in judgments of misinformation (e.g., Batailler et al., 2022; Gawronski et al., 2023), we used Signal Detection Theory (SDT; Green & Swets, 1966) to calculate indices for truth sensitivity and acceptance threshold. SDT’s index for truth sensitivity, d' , captures the extent to

which participants accurately distinguish between true and false information. High truth-sensitivity scores indicate a higher likelihood of responding “true” to true statements (i.e., *Hits*) and “false” to false statements (i.e., *Correct Rejections*). Conversely, low truth-sensitivity scores indicate a higher likelihood of responding “false” to true statements (i.e., *Misses*) and “true” to false statements (i.e., *False Alarms*). SDT’s index for acceptance threshold, c , captures participants’ general tendency to respond “false” (vs. “true”) irrespective of whether the statements are true or false. High acceptance-threshold scores indicate a higher likelihood of responding “false” to both false statements (i.e., *Correct Rejections*) and true statements (i.e., *Misses*). Conversely, low acceptance-threshold scores indicate a higher likelihood of responding “true” to both false statements (i.e., *False Alarms*) and true statements (i.e., *Hits*).

Following our preregistered data aggregation plan, hit and false-alarm rates were first calculated for pro-UK and pro-France statements. Hit rates (H) were calculated as the proportion of true statements judged as true. False-alarm rates (FA) were calculated as the proportion of false statements judged as true. Following recommendations by Macmillan and Creelman (2004), we converted values of 0 to $1/(2 \times N)$ and values of 1 to $1 - 1/(2 \times N)$, with N being the number of trials (i.e., 15 in our case). Truth-sensitivity scores were calculated using SDT’s equation for discrimination sensitivity: $d' = z(H) - z(FA)$. A truth sensitivity score of 0 reflects chance-level discernment. Acceptance-threshold scores were calculated using SDT’s equation for response bias: $c = -0.5 \times [z(H) + z(FA)]$.

Partisan bias is reflected in lower acceptance thresholds for identity-congenial compared to identity-uncongenial statements (Batailler et al., 2022; Gawronski et al., 2023; see also Ditto et al., 2025). The central question of the current study was whether partisan bias emerges for randomly assigned groups, in that participants show a lower acceptance threshold for statements that favor their randomly assigned ingroup compared to statements that favor their randomly assigned outgroup. Specifically, we tested the preregistered hypotheses that acceptance thresholds in judgments of pro-UK and pro-France information will differ as a function of group assignment, such that participants assigned to Team UK will show a lower threshold for pro-UK information than for pro-France information and participants assigned to Team France will show a lower threshold for pro-France information

¹ After completion of the main experiment, we were alerted that one pro-UK statement in our stimulus set was incorrectly classified as true, despite it being false (*The majority of French people find infidelity morally acceptable*). Deviating from our preregistered data aggregation plan, we did not include this statement in the aggregation

of truth judgments. The overall results do not differ depending on the exclusion of this item (see Supplemental Materials). We double-checked the classifications of all other statements to confirm they are correct.

than for pro-UK information. Acceptance thresholds in the No Team condition should not differ for pro-UK and pro-France information.

While some have argued that partisan bias should lead to lower truth discernment (or truth sensitivity) for identity-congenial compared to identity-uncongenial information (e.g., Pennycook & Rand, 2021b), a tendency to accept identity-congenial and to reject identity-uncongenial information should be reflected in differential acceptance thresholds, and identity effects on acceptance thresholds may not necessarily affect truth sensitivity (see Gawronski, 2021; Gawronski et al., 2025). For identity-congenial information, partisan bias involves not only a higher likelihood that false information is judged as true (i.e., increased false-alarm rate) but also a higher likelihood that true information is judged as true (i.e., increased hit rate), which should lead to an overall null effect on truth sensitivity. Conversely, for identity-uncongenial information, partisan bias involves not only a higher likelihood that true information is judged as false (i.e., reduced hit rate) but also a higher likelihood that false information is judged as false (i.e., reduced false-alarm rate), which should lead to an overall null effect on truth sensitivity. Based on these considerations, we further tested the preregistered hypotheses that group assignment will not affect truth sensitivity overall or differentially for pro-UK and pro-France information.

Results

Means and 95% confidence intervals for the number of statements accepted as true per condition are reported in Table 2.

Manipulation Check

To confirm the effectiveness of the group assignment, we calculated group-identification scores based on participants' responses to the six identification items. Following our preregistered data aggregation plan, group-identification scores were calculated by averaging responses to the three items for each country and then subtracting the mean identification scores for France from the mean identification scores for the UK. Thus, positive scores on this index indicate stronger relative identification with France, whereas negative scores indicate stronger relative identification with the UK. A preregistered one-way ANOVA comparing group-identification scores across Group Assignment conditions (Team UK vs. Team France vs. No Team) revealed a significant difference across conditions ($M_{\text{Team UK}} = -1.20$ vs. $M_{\text{Team France}} = 0.25$ vs. $M_{\text{No Team}} = -0.41$), $F(2, 560) = 41.86$, $p < .001$, $\eta_p^2 = .13$, 95% CI [.08, .18]. Preregistered post hoc t -tests confirmed the effectiveness of the group assignment, supporting all preregistered hypotheses. Specifically, relative identification with France vs. UK was significantly stronger in the Team France condition compared to the Team UK condition, $t(373) = 8.00$, $p < .001$, $d = 0.83$,

95% CI [0.62, 1.04], and the No Team condition, $t(371) = 4.45$, $p < .001$, $d = 0.46$, 95% CI [0.25, 0.67]. In addition, relative identification with France vs. UK was significantly stronger in the No Team condition compared to the Team UK condition, $t(376) = 5.48$, $p < .001$, $d = 0.56$, 95% CI [0.36, 0.77].

Acceptance Thresholds

In our primary analysis concerning partisan bias, we submitted acceptance-threshold scores to a preregistered 3 (Group Assignment: Team UK vs. Team France vs. No Team) $\times$ 2 (Statement Slant: pro-UK vs. pro-France) mixed ANOVA with the first factor varying between-subjects and the second factor varying within-subjects. The main effects of Group Assignment, $F(2, 560) = 0.28$, $p = .756$, $\eta_p^2 < .01$, 95% CI [.00, .01] and Statement Slant, $F(1, 560) = 2.91$, $p = .088$, $\eta_p^2 = .01$, 95% CI [.00, .02] were not significant. However, as predicted, there was a significant interaction between Group Assignment and Statement Slant, $F(2, 560) = 8.79$, $p < .001$, $\eta_p^2 = .03$, 95% CI [.01, .06] (see Fig. 1). Following our preregistered analysis plan, we conducted follow-up t -tests for dependent means to examine this interaction. As predicted, acceptance thresholds in the Team UK condition were significantly lower for pro-UK statements compared to pro-France statements, $t(189) = -4.02$, $p < .001$, $d = -0.38$, 95% CI [-0.57, -0.19]. While acceptance thresholds in the Team France condition were descriptively lower for pro-France statements compared to pro-UK statements, the observed difference did not reach statistical significance, $t(184) = -1.35$, $p = .179$, $d = -0.12$, 95% CI [-0.30, 0.06]. Finally, as predicted, acceptance thresholds in the No Team condition did not significantly differ for pro-UK statements and pro-France statements, $t(187) = -0.01$, $p = .991$, $d < 0.01$, 95% CI [-0.18, 0.18].

Non-preregistered follow-up t -tests comparing acceptance thresholds in the control versus experimental conditions separately by statement type (pro-UK vs. pro-France) revealed a significant difference between the Team UK and the No Team conditions for pro-UK statements, $t(376) = -2.48$, $p = .014$, $d = -0.26$, 95% CI [-0.46, -0.05]. All other comparisons for pro-UK statements ($t(371) = 0.49$; $p = .624$) and pro-France statements (all t s < 1.42 ; all p s $\geq .158$) were not significant. We also conducted non-preregistered follow-up t -tests to compare partisan bias in the Team UK and Team France conditions to the No Team condition, respectively. To do so, we calculated partisan-bias scores by subtracting acceptance thresholds for pro-France statements from acceptance thresholds for pro-UK statements ($c_{\text{pro-UK}} - c_{\text{pro-France}}$). Positive scores on this index reflect partisan bias favoring France over the UK, whereas negative scores reflect partisan bias favoring the UK over France. Analyses revealed that partisan bias favoring the UK over France was significantly greater in the Team UK condition

compared to the No Team condition, $t(376) = -3.06, p = .002, d = -0.31, 95\% \text{ CI } [-0.52, -0.11]$. However, partisan-bias scores did not significantly differ between the Team France condition and the No Team condition, $t(371) = 1.01, p = .313, d = 0.10, 95\% \text{ CI } [-0.10, 0.31]$.

Truth Sensitivity

Because truth sensitivity and partisan bias are independent (Gawronski 2021; Gawronski et al., 2025), we predicted that truth-sensitivity scores would be unaffected by the group-assignment manipulation. To test this hypothesis, we submitted truth-sensitivity scores to a preregistered 3 (Group Assignment: Team UK vs. Team France vs. No Team) $\times$ 2 (Statement Slant: pro-UK vs. pro-France) mixed ANOVA with the first factor varying between-subjects and the second factor varying within-subjects. Results supported all preregistered hypotheses, in that the main effect of Group Assignment was not significant, $F(2, 560) = 0.71, p = .495, \eta_p^2 < .01, 95\% \text{ CI } [.00, .01]$, nor was the interaction between Group Assignment and Statement Slant, $F(2, 560) = 0.88, p = .417, \eta_p^2 < .01, 95\% \text{ CI } [.00, .02]$ (see Fig. 2). Furthermore, across conditions, truth-sensitivity scores were significantly greater than zero, $t(562) = 40.13, p < .001, d = 1.69, 95\% \text{ CI } [1.56, 1.82]$, and comparable in size to truth-sensitivity scores found in research on political misinformation (e.g., Gawronski et al., 2023), indicating that participants did not randomly guess the truth or falsity of the presented statements.

Discussion

Experiment 1 provides initial evidence that partisan bias in judgments of true and false information can emerge for randomly assigned identities without differences in prior knowledge. However, while participants in the Team UK condition showed a significant partisan-bias effect, the predicted partisan-bias effect in the Team France condition was not statistically significant. Upon further inspection, this asymmetry may have been due to an unexpected baseline preference for the UK over France, as reflected in the finding that participants in the No Team condition showed a significantly stronger identification with the UK compared to France.² This baseline asymmetry may have suppressed the induced identification with France in the Team France condition, reducing the strength of the predicted partisan-bias effect.³ Expanding on these considerations, the two goals of Experiment 2 were to (i) replicate the main findings of Experiment 1 with a larger sample, and (ii) resolve the identified asymmetry by using countries that showed equal baseline identification scores in a pilot study (i.e., Spain vs. Greece).

Experiment 2

Method

Participants and Design

Experiment 2 utilized a 3 (Group Assignment: Team Spain vs. Team Greece vs. No Team) $\times$ 2 (Statement Accuracy: true vs. false) $\times$ 2 (Statement Slant: pro-Spain vs. pro-Greece) mixed design with the first factor varying between-subjects and the other two factors varying within-subjects. We aimed to recruit a total of 900 participants. Sensitivity analyses using GPower 3.1.9.6 (Faul et al., 2007) demonstrated that a sample of 900 provides 80% power to detect a small effect of $f = 0.061$ for the predicted interaction between Group Assignment and Statement Slant, assuming a correlation between measures of $r = .30$ and using a nonsphericity correction of $\epsilon = 1$.

Participants were recruited via Prolific Academic. Participation was restricted to Prolific workers who (1) are U.S. citizens currently residing in the U.S., (2) are fluent in English, (3) have completed at least 100 previous assignments, (4) had an approval rating of 95% or higher on previous submissions, and (5) did not participate in Experiment 1. Completion of the study took approximately 15 minutes and participants were compensated with US-\$3.

A total of 907 participants completed the study (seven participants completed the study but did not request compensation). After excluding participants who failed the preregistered attention check ($n_{\text{Team Spain}} = 18, n_{\text{Team Greece}} = 19, n_{\text{No Team}} = 22$; Oppenheimer et al., 2009), our final sample consisted of 848 participants (283 Team Spain; 284 Team Greece; 281 No Team). Exclusions did not significantly differ across conditions, $\chi^2(2) = 0.45, p = .799$. Demographic information is provided in Table 1.

Procedure

Experiment 2 used the same procedure as Experiment 1, the only difference being that we used Spain and Greece as countries for the group-assignment manipulation. We also slightly modified the instructions for the group assignment to better align them with the relation between the two countries. Spain and Greece were selected based on the results of pilot testing, which revealed equal identification with the two countries among U.S. participants (for more details, see Supplemental Materials). Participants judged the truth of 60 statements that varied in terms of their veracity (true vs. false) and slant (pro-Spain vs. pro-Greece). Examples of the statements used include: *Spain has produced more Nobel Prize winners than Greece* (pro-Spain true) and

² A t -test for dependent means revealed that, in the No Team condition, identification with the UK was significantly stronger than identification with France, $t(187) = 5.86, p < .001, d = 0.36, 95\% \text{ CI } [0.24, 0.48]$.

³ Consistent with this post-hoc assumption, participants in the Team UK condition identified more strongly with the UK than France, $t(189) = 9.61, p < .001, d = 0.95, 95\% \text{ CI } [0.72, 1.18]$, while the reverse effect in the Team France condition was only marginal, $t(184) = -1.88, p = .062, d = -0.18, 95\% \text{ CI } [-0.38, 0.01]$.

Greece was ranked as a more charitable country than Spain (pro-Greece false). Statements were pilot tested to ensure that the slant of the statements matched the respective condition and that the average slant was of equal strength across conditions. For further details on the selection and pilot testing of statements, see Supplemental Materials. Data were analyzed using SDT following the procedures in Experiment 1.

Results

Means and 95% confidence intervals for the number of statements accepted as true per condition are reported in Table 3.

Manipulation Check

Group-identification scores were calculated following the procedures in Experiment 1 such that higher scores represent stronger identification with Greece compared to Spain. A preregistered one-way ANOVA comparing group-identification scores across Group Assignment conditions (Team Spain vs. Team Greece vs. No Team) revealed a significant difference across conditions ($M_{\text{Team Spain}} = -0.81$ vs. $M_{\text{Team Greece}} = 0.98$ vs. $M_{\text{No Team}} = -0.02$), $F(2, 845) = 100.21$, $p < .001$, $\eta_p^2 = 0.19$, 95% CI [.15, .24]. Preregistered post-hoc t -tests confirmed the effectiveness of the group assignment, supporting all preregistered hypotheses. Specifically, relative identification with Greece vs. Spain was significantly stronger in the Team Greece condition compared to the Team Spain condition, $t(565) = 12.44$, $p < .001$, $d = 1.04$, 95% CI [0.87, 1.22], and the No Team condition, $t(563) = 8.47$, $p < .001$, $d = 0.71$, 95% CI [0.54, 0.88]. Moreover, relative identification with Greece vs. Spain was significantly stronger in the No Team condition compared to the Team Spain condition, $t(562) = 6.75$, $p < .001$, $d = 0.57$, 95% CI [0.40, 0.74]. Further confirming the effectiveness of the group assignment manipulation, non-preregistered t -tests for dependent means revealed that participants in the Team Spain condition identified more strongly with Spain than Greece, $t(282) = 8.05$, $p < .001$, $d = 0.66$, 95% CI [0.48, 0.84]; participants in the Team Greece condition identified more strongly with Greece than Spain, $t(283) = 9.53$, $p < .001$, $d = 0.79$, 95% CI [0.60, 0.97]; and participants in the No Team condition did not significantly differ in their identification with Spain and Greece, $t(280) = 0.39$, $p = .699$, $d = 0.02$, 95% CI [-0.09, 0.13].

Acceptance Thresholds

In our primary analysis concerning partisan bias, we submitted acceptance-threshold scores to a preregistered 3 (Group Assignment: Team Spain vs. Team Greece vs. No Team) $\times$ 2 (Statement Slant: pro-Spain vs. pro-Greece) mixed ANOVA with the first factor varying between-subjects and the second factor varying within-subjects. The main effects of Group Assignment, $F(2, 845) = 1.01$, $p = .364$, $\eta_p^2 < .01$, 95% CI [.00, .01], and Statement Slant, $F(1, 845) = 2.26$, $p = .133$, $\eta_p^2 < .01$,

95% CI [.00, .01], were not significant. However, as predicted, there was a significant interaction between Group Assignment and Statement Slant, $F(2, 845) = 10.80$, $p < .001$, $\eta_p^2 = .02$, 95% CI [.01, .05] (see Fig. 3). Following our preregistered analysis plan, we conducted follow-up t -tests to examine this interaction. All preregistered hypotheses were supported. Specifically, acceptance thresholds in the Team Spain condition were significantly lower for pro-Spain statements compared to pro-Greece statements, $t(282) = -2.46$, $p = .015$, $d = -0.17$, 95% CI [-0.30, -0.03]. Acceptance thresholds in the Team Greece condition were significantly lower for pro-Greece compared to pro-Spain statements, $t(283) = -3.71$, $p < .001$, $d = -0.26$, 95% CI [-0.40, -0.12]. Finally, acceptance thresholds in the No Team condition did not differ significantly for pro-Spain versus pro-Greece statements, $t(280) = 1.17$, $p = .242$, $d = 0.07$, 95% CI [-0.05, 0.19].

Non-preregistered follow-up t -tests comparing acceptance thresholds in the control versus experimental conditions separately by statement type (pro-Spain vs. pro-Greece) revealed a significant difference between the Team Spain and the No Team conditions for pro-Greece statements, $t(562) = 2.29$, $p = .023$, $d = 0.19$, 95% CI [0.03, 0.36]. All other comparisons for pro-Spain statements (all t s < 1.00 ; all p s $> .31$) and pro-Greece statements ($t(563) = -1.43$; $p = .154$) were not significant. We also conducted non-preregistered follow-up t -tests to compare partisan bias in the Team Spain and Team Greece conditions to the No Team condition, respectively. To do so, we calculated partisan-bias scores by subtracting acceptance thresholds for pro-Greece statements from acceptance thresholds for pro-Spain statements ($c_{\text{pro-Spain}} - c_{\text{pro-Greece}}$). Positive scores on this index reflect partisan bias favoring Greece over Spain, whereas negative scores reflect partisan bias favoring Spain over Greece. Analyses revealed that partisan bias favoring Spain over Greece was significantly greater in the Team Spain condition compared to the No Team condition, $t(562) = -2.63$, $p = .009$, $d = -0.22$, 95% CI [-0.39, -0.06], and that partisan bias favoring Greece over Spain was significantly greater in the Team Greece condition compared to the No Team condition, $t(563) = 2.16$, $p = .031$, $d = 0.18$, 95% CI [0.02, 0.35].

Truth Sensitivity

As in Experiment 1, we predicted that truth-sensitivity scores would be unaffected by the group-assignment manipulation. To test this hypothesis, we conducted a preregistered 3 (Group Assignment: Team Spain vs. Team Greece vs. No Team) $\times$ 2 (Statement Slant: pro-Spain vs. pro-Greece) mixed ANOVA with the first factor varying between-subjects and the second factor varying within-subjects. Results supported all preregistered hypotheses, in that neither the main effect of Group Assignment, $F(2, 845) = 0.31$, $p = .733$, $\eta_p^2 <$

.01, 95% CI [.00, .01], nor the interaction between Group Assignment and Statement Slant, $F(2, 845) = 1.84$, $p = 0.160$, $\eta_p^2 < .01$, 95% CI [.00, .02], were statistically significant (see Fig. 4). Across conditions, truth-sensitivity scores were significantly greater than zero, $t(847) = 40.41$, $p < .001$, $d = 1.39$, 95% CI [1.29, 1.48], and again comparable in size to truth-sensitivity scores found in research on political misinformation (e.g., Gawronski et al., 2023), indicating that participants did not randomly guess the truth or falsity of the presented statements.

General Discussion

Across two experiments, we found that participants were more accepting of information that was congenial versus uncongenial to a randomly assigned identity. Crucially, because assignment to the groups was random, the observed partisan-bias effect could not be driven by differential knowledge of identity-congenial versus identity-uncongenial information. Hence, the current findings support the hypothesized role of identity protection as a factor underlying partisan bias in judgments of true and false information (see Van Bavel et al., 2024).

While the use of randomly assigned groups allowed us to more cleanly rule out effects of differential knowledge, a natural next question is: how much does identity-protection contribute to partisan bias relative to knowledge-based effects? While the effect sizes for partisan bias in the current experiments were relatively small (η_p^2 's = .03), they plausibly represent lower bounds on the effects of identity-protection, given that motivational effects of randomly-assigned identities are likely much weaker compared to those of existing identities with personal significance (e.g., Democrat versus Republican). By modeling the effects of preferences and prior expectations, Celniker and Ditto (2024) provided evidence that effects of motivated reasoning and differential knowledge are similar in size. However, future research may quantify the relative contribution of identity protection and differential knowledge in judgments of misinformation.

Another important question concerns the mechanisms underlying the obtained effects. Research on motivated reasoning often pits “hot” against “cold” processes. However, randomly assigning participants to national teams seems unlikely to evoke strong affective responses, raising questions about alternative mechanisms. Critics might argue that, in the absence of relevant knowledge, participants may have used their assigned group-identity as a default heuristic to guide their judgments. However, it is not the case that participants in the current studies had no relevant knowledge. Effect sizes for truth sensitivity were quite large with Cohen's d s of 1.7 (Experiment 1) and 1.4 (Experiment 2), indicating that participants had

substantial knowledge. Hence, it seems unlikely that participants were merely relying on their assigned groups as a default heuristic to compensate for a lack of relevant knowledge. Future studies are needed to investigate the mechanisms underlying identity-protection effects on judgments of true and false information.

While the current work supports the role of identity-protective motivated reasoning in the acceptance of misinformation, it is based on a conceptualization of motivated reasoning as the tendency to arrive at conclusions one wants to arrive at (Kunda, 1990). The findings do not speak to an alternative conceptualization of motivated reasoning as enhanced partisan bias resulting from cognitive elaboration (motivated-system-2 hypothesis; see Kahan, 2013; Pennycook & Rand, 2021a). In terms of the latter conceptualization, the difference in acceptance thresholds for identity-congenial versus identity-uncongenial information should increase with greater cognitive elaboration (see Batailler et al., 2022). Future research may measure or manipulate cognitive elaboration to test the validity of the motivated-system-2 hypothesis (for related evidence, see Sultan et al., 2024).

Overall, the current findings have important implications for addressing partisan bias in the acceptance of misinformation (Gawronski et al., 2024). While it is often implied that acceptance of misinformation is due to the other side not “having the facts”, our results demonstrate that this is not the full story. Even if partisans have identical knowledge about the relevant facts, they may still come to opposing conclusions. Thus, while it remains important to reduce knowledge differences arising from echo chambers and selective exposure, the current research highlights that an exclusive focus on knowledge-based effects will be insufficient to eliminate partisan bias. Instead, future interventions aimed at reducing partisan bias must be designed to also target people's underlying motivations (Hornsey & Fielding, 2017).

Conclusion

The current experiments provide a novel test addressing the debate between identity-protection and differential-knowledge accounts of partisan bias in judgments of misinformation. By utilizing randomly assigned identities, as opposed to groups with pre-existing differences in prior knowledge, we showed that partisan bias occurs even in the absence of differential knowledge. These findings demonstrate that misinformation susceptibility is not merely due to differences in knowledge, but also to underlying motivations. Thus, to prevent the acceptance of misinformation, interventions must be designed to consider the motivations underlying partisan bias in judgments of true and false information.

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Research Transparency Statement

General Disclosures

Conflicts of interest: All authors declare no conflicts of interest.

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Computational reproducibility: The computational reproducibility of the results in the main article (but not the supplementary materials) has been independently confirmed by the journal's STAR team.

Experiment 1 Disclosures

Preregistration: The hypotheses, methods, and analyses were preregistered (<https://osf.io/kpcfw>) prior to data collection. There was one deviation from the preregistration. One item of the truth judgment task was not included in the aggregation of data due to an incorrect truth classification in the preregistration. All original analyses as preregistered are reported in the supplemental materials.

Materials: All materials are publicly available (<https://osf.io/gj8wh/>).

Data: All primary data are publicly available (<https://osf.io/gj8wh/>).

Analysis scripts: All analysis scripts are publicly available (<https://osf.io/gj8wh/>).

Experiment 2 Disclosures

Preregistration: The hypotheses, methods, and analyses were preregistered (<https://osf.io/jcp8q>) prior to data collection. There were no deviations from the preregistration.

Materials: All materials are publicly available (<https://osf.io/gj8wh/>).

Data: All primary data are publicly available (<https://osf.io/gj8wh/>).

Analysis scripts: All analysis scripts are publicly available (<https://osf.io/gj8wh/>).

Table 1*Participant Demographics in Experiments 1 and 2.*

	Experiment 1	Experiment 2
<i>Age – M (SD)</i>	38.95 (12.11)	41.66 (13.61)
<i>Gender</i>		
Female	335	463
Male	217	376
Prefer not to say	1	2
Other	10	7
<i>Race</i>		
Asian	33	40
American Indian/Alaska Native	3	6
Black	96	130
Native Hawaiian or Pacific Islander	2	1
White	396	630
Mixed	26	29
Other	7	12
<i>Education</i>		
Less than high school degree	2	6
High school degree	62	94
Some college but no degree	122	164
Associate degree	52	85
Bachelor's degree	243	338
Master's degree	62	139
Doctoral or professional degree	20	22

Table 2

Means and 95% Confidence Intervals of the Number of Statements Accepted as True as a Function of Group Assignment, Experiment 1.

Group	Pro-UK				Pro-France			
	True		False		True		False	
	<i>M</i>	95% CI	<i>M</i>	95% CI	<i>M</i>	95% CI	<i>M</i>	95% CI
Team UK	9.81	[9.45, 10.15]	6.56	[6.12, 6.99]	9.74	[9.39, 10.10]	5.63	[5.24, 6.01]
Team France	9.42	[9.04, 9.79]	5.61	[5.23, 5.99]	10.18	[9.82, 10.55]	6.06	[5.69, 6.44]
No Team	9.48	[9.16, 9.80]	5.78	[5.42, 6.14]	9.98	[9.65, 10.31]	6.01	[5.64, 6.37]

Note. Each cell includes 15 statements, except for Pro-UK True which includes 14 statements (see Footnote 1).

Table 3

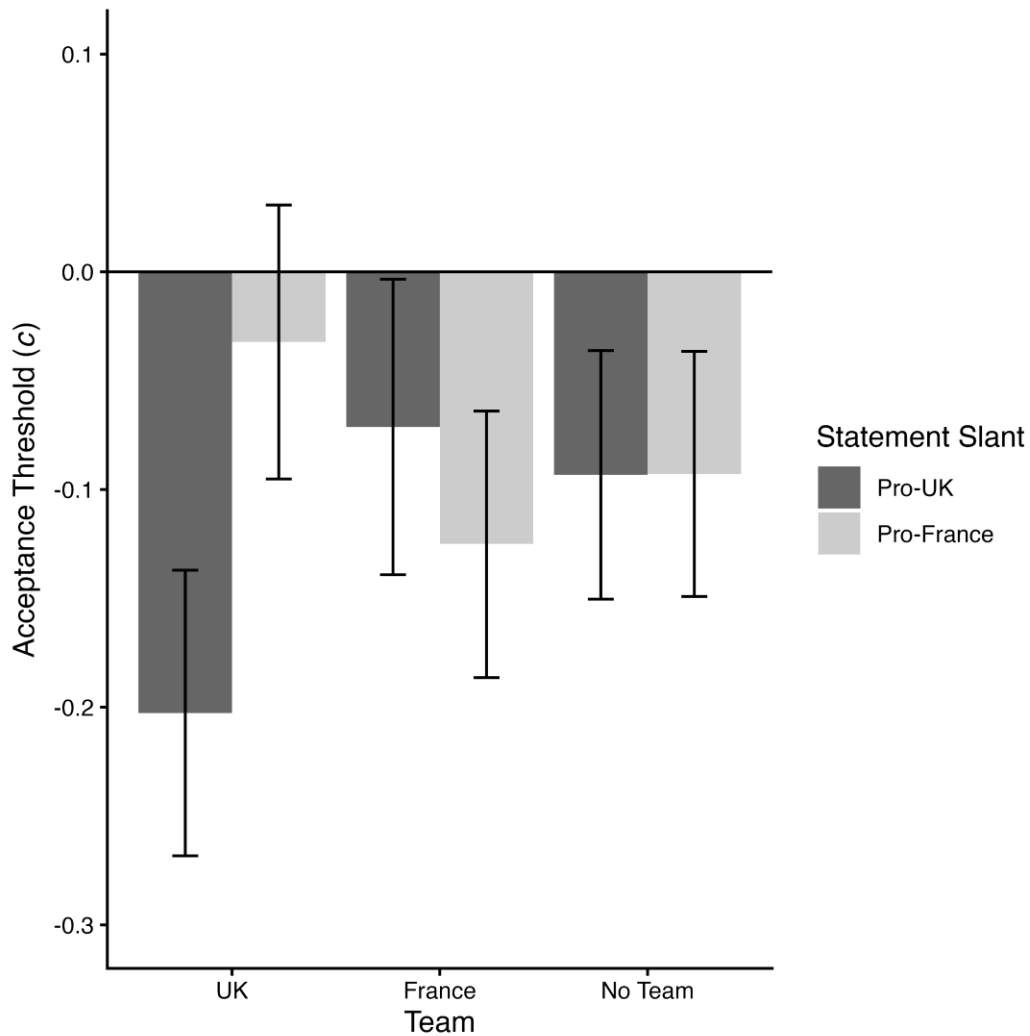
Means and 95% Confidence Intervals of the Number of Statements Accepted as True as a Function of Group Assignment, Experiment 2.

Group	Pro-Spain				Pro-Greece			
	True		False		True		False	
	<i>M</i>	95% CI	<i>M</i>	95% CI	<i>M</i>	95% CI	<i>M</i>	95% CI
Team Spain	8.81	[8.50, 9.13]	5.85	[5.54, 6.15]	8.67	[8.36, 8.98]	5.32	[4.96, 5.68]
Team Greece	8.63	[8.32, 8.95]	5.42	[5.10, 5.75]	9.48	[9.19, 9.76]	6.00	[5.62, 6.37]
No Team	8.89	[8.58, 9.20]	5.56	[5.24, 5.87]	9.02	[8.74, 9.31]	5.82	[5.45, 6.18]

Note. Each cell includes a total of 15 statements.

Figure 1

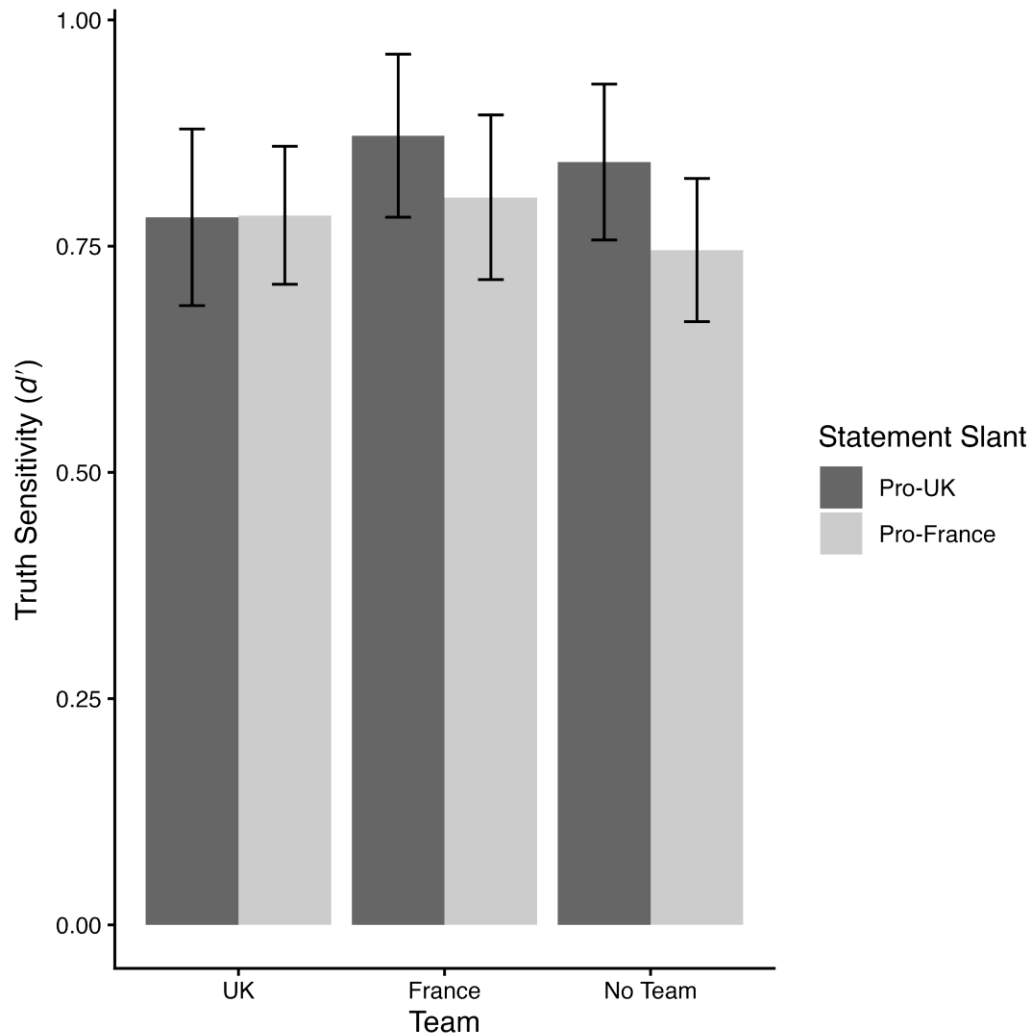
Acceptance Threshold as a Function of Group Assignment and Statement Slant, Experiment 1.



Note. Acceptance threshold scores (c) were calculated using signal detection theory. Lower scores reflect a stronger tendency to accept information. Error bars represent 95% confidence intervals.

Figure 2

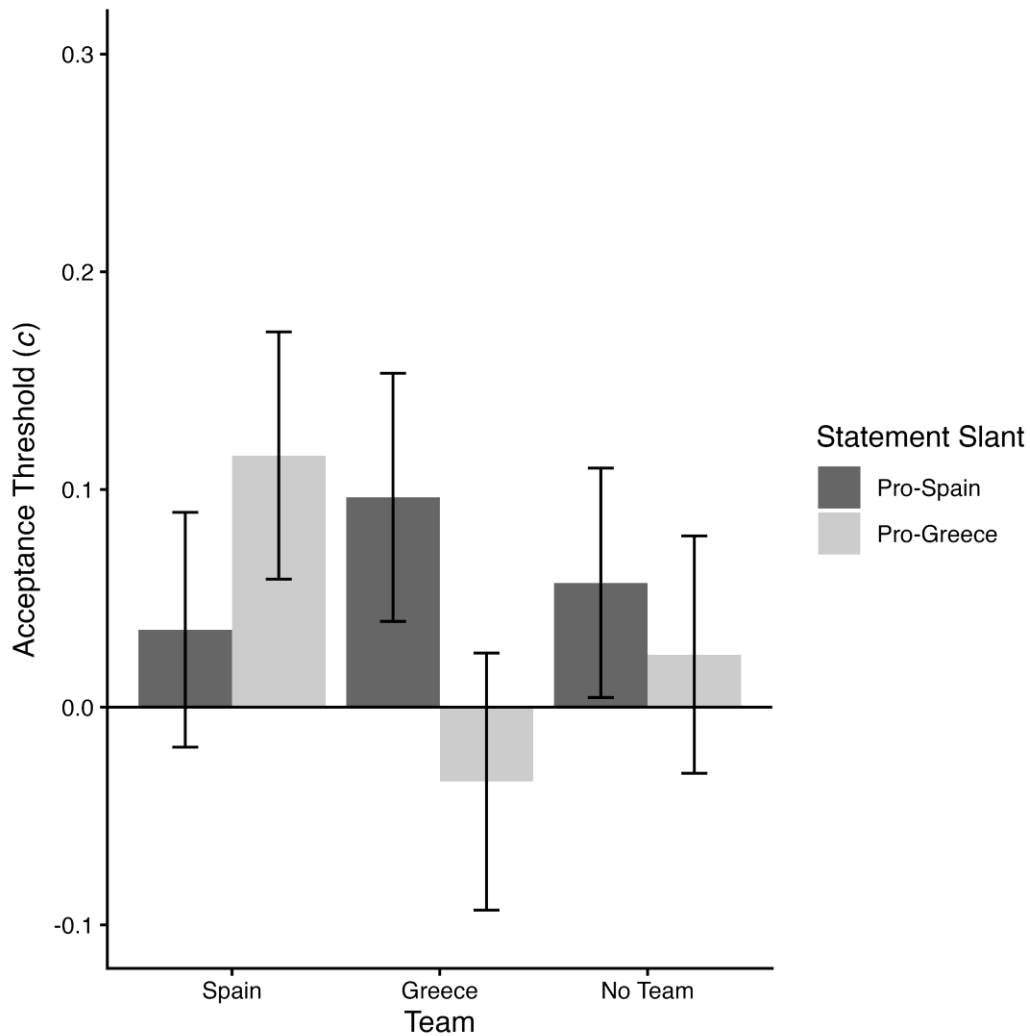
Truth Sensitivity as a Function of Group Assignment and Statement Slant, Experiment 1.



Note. Truth sensitivity scores (d') were calculated using signal detection theory. Higher scores reflect greater accuracy in discerning true from false information. Error bars represent 95% confidence intervals.

Figure 3

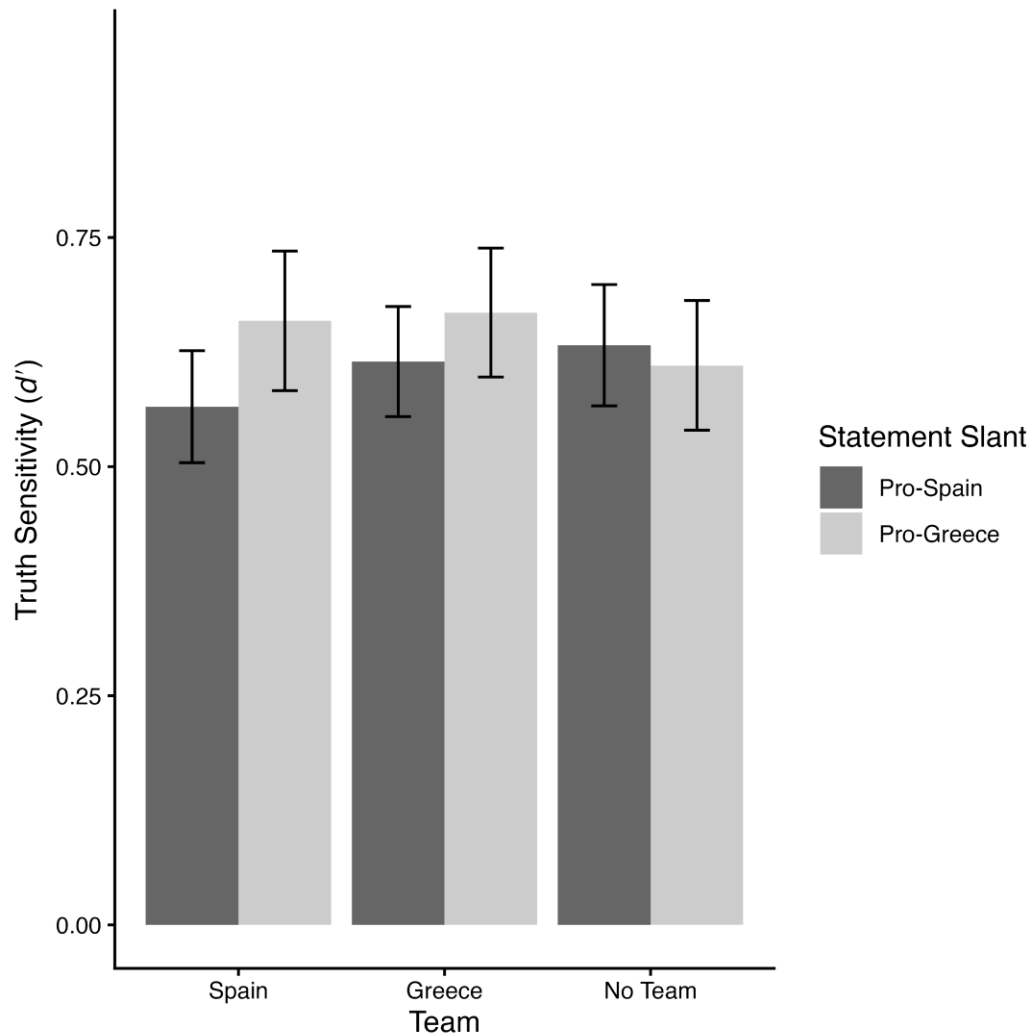
Acceptance Threshold as a Function of Group Assignment and Statement Slant, Experiment 2.



Note. Acceptance threshold scores (c) were calculated using signal detection theory. Lower scores reflect a stronger tendency to accept information. Error bars represent 95% confidence intervals.

Figure 4

Truth Sensitivity as a Function of Group Assignment and Statement Slant, Experiment 2.



Note. Truth sensitivity scores (d') were calculated using signal detection theory. Higher scores reflect greater accuracy in discerning true from false information. Error bars represent 95% confidence intervals.